

No. 25-13631-C

**In The United States Court of Appeals
for the Eleventh Circuit**

CITADEL SECURITIES LLC

Petitioner,

v.

U.S. SECURITIES AND EXCHANGE COMMISSION

Respondent.

INVESTORS EXCHANGE, LLC

Intervenor.

On Petition for Review of a Final Order of the
United States Securities and Exchange Commission

**BRIEF OF CTC, LLC AS *AMICUS CURIAE*
IN SUPPORT OF RESPONDENT**

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CERTIFICATE OF INTERESTED PERSONS

Pursuant to Federal Rule of Appellate Procedure 26.1 and 11th Circuit Rule 26.1-1, *Amicus* CTC, LLC provides the following certificate of interested persons:

1. Bradley, Keith, counsel for *Amicus Curiae* Prof. J.W. Verrett;
2. Burns, Thomas A., counsel for *Amicus Curiae* Healthy Markets Association;
3. Citadel Securities GP LLC, indirect, ultimate controlling entity of Citadel Securities LLC;
4. Citadel Securities LLC, Petitioner;
5. CTC, LLC, *Amicus Curiae* supporting Respondent Securities and Exchange Commission;
6. Dentons US LLP, counsel for *Amici Curiae* NYSE Arca, Inc., NYSE American LLC, and MEMX LLC;
7. Doumani, Tala, counsel for Intervenor Investors Exchange LLC;
8. Gibson, Dunn & Crutcher LLP, counsel for Petitioner Citadel Securities LLC;
9. Glickstein, Jed W., counsel for *Amicus Curiae* CTC, LLC;
10. Healthy Markets Association, *Amicus Curiae* supporting Respondent Securities and Exchange Commission;
11. Henkin, Douglas W., counsel for *Amici Curiae* NYSE Arca, Inc., NYSE American LLC, and MEMX LLC;
12. Kaskel, Jonathan H, counsel for *Amici Curiae* NYSE Arca, Inc., NYSE American LLC, and MEMX LLC;
13. Investors Exchange LLC, Intervenor;

14. MEMX LLC, *Amicus Curiae* supporting Petitioner Citadel Securities LLC;
15. NYSE American LLC, *Amicus Curiae* supporting Petitioner Citadel Securities LLC;
16. NYSE Arca, Inc., *Amicus Curiae* supporting Petitioner Citadel Securities LLC;
17. Parise, Emily True, counsel for Respondent Securities and Exchange Commission;
18. Richman, Brian A., counsel for Petitioner Citadel Securities LLC;
19. Savitt, William, counsel for Intervenor Investors Exchange LLC;
20. Scalia, Eugene, counsel for Petitioner Citadel Securities LLC;
21. Securities and Exchange Commission, Respondent;
22. Shin, Won S., counsel for Intervenor Investors Exchange LLC;
23. Tienken, John W., counsel for Petitioner Citadel Securities LLC;
24. Verrett, Prof. J.W., *Amicus Curiae* supporting Respondent Securities and Exchange Commission;
25. Wachtell, Lipton, Rosen & Katz, counsel for Intervenor Investors Exchange LLC;
26. Wagner, Brooke, counsel for Respondent Securities and Exchange Commission;
27. Walker, Helgi C., counsel for Petitioner Citadel Securities LLC; and,
28. Wolf, Brandon, counsel for Petitioner Citadel Securities LLC.

To the undersigned's knowledge, no publicly traded company or corporation has an interest in the outcome of the case. CTC, LLC will file an amended certificate of interested persons should it become aware of a change in interests that would affect

the disclosures required by Federal Rule of Appellate Procedure 26.1 and 11th Circuit Rule 26.1-4.

CORPORATE DISCLOSURE STATEMENT

Pursuant to Federal Rule of Appellate Procedure 26.1 and 11th Circuit Rule 26.1-2, *Amicus* CTC, LLC states that it is a wholly owned subsidiary of CTC Trading Group, LLC, which is privately held, and that no publicly held corporation holds 10% or more of its stock.

Dated: February 6, 2026

Respectfully submitted,

/s/ Jed W. Glickstein

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INTEREST OF *AMICUS CURIAE*¹

Chicago Trading Company (CTC) is a global proprietary trading firm providing liquidity in regulated futures and securities markets in the United States and internationally. CTC, LLC, is a registered broker-dealer and market maker in U.S. equity and index options and a member of Cboe, C2, BZX, EDGX, NYSE Arca, NYSE American, Nasdaq ISE, Nasdaq Phlx, and FINRA. As an active participant in the options markets and a firm that will be directly affected by the outcome of this case, CTC has a significant interest in the proper resolution of the question presented.

STATEMENT OF THE ISSUE

Whether the Securities and Exchange Commission acted reasonably in approving the proposed rule change to adopt rules to govern the trading of options on IEX Options LLC, as amended, as consistent with the Securities Exchange Act of 1934, 15 U.S.C. § 78f(b), and the rules and regulations thereunder applicable to a national securities exchange.

SUMMARY OF ARGUMENT

The Securities and Exchange Commission's approval of the IEX Options proposal—including a hardware-based latency mechanism that would add a *de*

¹ Pursuant to Federal Rule of Appellate Procedure 29(a)(2), CTC states that no counsel for a party authored this brief in whole or in part and no person other than CTC or its counsel made a monetary contribution to the preparation or submission of this brief. All parties (Petitioner, Respondent, and Intervenor) have consented to the filing of this brief.

minimis delay of 0.000350 seconds to each incoming order and quote message—was lawful, well-reasoned, and fully consistent with the Securities Exchange Act’s requirements. *See generally* 15 U.S.C. § 78f(b). CTC submits this brief to provide the Court with a practitioner’s perspective on the following points.

First, as the Commission correctly recognized, the IEX Options proposal is procompetitive. Market makers like CTC are uniquely exposed to the risk of latency arbitrage, which creates a wasteful arms-race dynamic that results in wider spreads and less liquidity for investors. Using an exchange-controlled mechanism to mitigate this “pick-off” risk is an important innovation that will lower technological barriers to entry and allow more and smaller firms to compete at providing liquidity.

Second, the IEX Options proposal is not unfairly discriminatory. To the contrary, the proposal would remove unfair distortions in the market, encouraging new entrants to provide liquidity and promoting innovations in market structure. Claims that the rule “unfairly discriminates” against market participants in favor of IEX’s members ignore a fundamental reality of modern electronic markets—namely, that the risk of missing a displayed quote has always existed, and will always exist. IEX options quotes would not be “stale” or “illusory” in any new or unique way, and certainly not to any material degree given the extremely brief duration of the speedbump. The Commission’s approval of a similar speedbump for

equities demonstrates that the introduction of *de minimis* intentional latency is fully consistent with the Exchange Act and the operation of fair and orderly markets.

Petitioner Citadel Securities LLC's arguments to the contrary mischaracterize how the options markets function, ignore existing exchange mechanisms, and misconstrue the regulatory framework and precedent. CTC urges the Court to deny the petition for review.

ARGUMENT

Founded in 1995, CTC is a proprietary trading firm specializing in options market making, systematic strategies, and cutting-edge trading technology. Options are derivative instruments that grant holders the right to buy or sell shares of equity at a specific date (the "expiration date") and price (the "strike price"). Options market makers supply liquidity by quoting prices to buy and sell in over one million options series every day. Like all market makers, CTC must consider the risk of latency arbitrage in designing its systems and deciding how aggressively to quote.²

In 2025, Investors Exchange LLC ("IEX") submitted a request for approval of rules to govern a new "IEX Options" market. As relevant here, the proposal included a hardware-based latency mechanism, sometimes referred to as a

² An "options series" is a group of options that share the same underlying security, strike price, and expiration date; an "options class" refers to the group of all options series with the same underlying stock. *SEC v. Wang*, 944 F.2d 80, 83 (2d Cir. 1991).

“speedbump,” that would add a *de minimis* delay of 0.000350 seconds to each incoming order and quote message on the exchange. The purpose of the speedbump was to enable IEX to obtain the most accurate view of the national market prior to processing quotes and orders. The proposal also included an accompanying Options Risk Parameter (“ORP”) that would permit IEX to cancel or reprice quotes in the event of significant changes in the price of the underlying during this 350 microsecond period. The ORP was designed to protect IEX market makers from excessive execution of quotes at stale prices (*e.g.*, before the market maker can update them in response to rapidly-changing market conditions).

Citadel Securities LLC (“Citadel”), joined by a group of exchanges (the “*Amici* Exchanges”), have asked this Court to set aside the Commission’s order approving IEX’s proposal. As explained below, however, the speedbump and ORP are procompetitive, innovative, far from unprecedented, and entirely consistent with the Exchange Act. A speedbump may mean more competition for large incumbents and among exchanges, but it will result in better outcomes for the market as a whole. The petition for review of the Commission’s order should be denied.

I. THE IEX PROPOSAL IS PROCOMPETITIVE AND WOULD BENEFIT INVESTORS.

Citadel asserts that the Commission failed to consider the supposed burdens the IEX proposal would impose on competition. According to Citadel, the Commission needed to “examine the conditions of competition in the options

market.” Br. 56–61. The *Amici* Exchanges argue likewise, stating that the Commission “the Order presumed it was necessary and appropriate to disrupt [a] functioning, competitive market without any evidence of need or market failure, and while ignoring significant concerns expressed by commenters about the implications of this novel mechanism.” Exchanges Br. 10–13. Both Citadel and the *Amici* Exchanges ignore that—as the Commission expressly found—latency arbitrage is an issue for many options market makers, and as a result the IEX proposal would likely be *procompetitive*, addressing a “legitimate disadvantage in latency arbitrage” faced by market makers that “lack the expensive low-latency systems, connectivity, and data sources used by those engaged in such strategies.” *Order Approving a Proposed Rule Change, as Modified by Amendment No. 3, To Adopt Rules To Govern the Trading of Options on the Exchange for a New Facility Called IEX Options*, 90 Fed. Reg. 45861, 45872 (Sept. 23, 2025).

The conclusion that the IEX proposal would not unduly burden—and would in fact promote—competition was rationally based on the facts before the Commission, *see Lindeen v. SEC*, 825 F.3d 646, 658 (D.C. Cir. 2016), and was entirely appropriate. To see why, it is useful to understand the problem of latency arbitrage from the market maker’s perspective.

A. Latency arbitrage induces a socially wasteful arms race for market makers.

Market makers like CTC serve a crucial function in modern financial markets. Their role is particularly important in less-liquid instruments, including many series of options, where the likelihood of natural buyers and sellers posting simultaneous offsetting orders would otherwise be very low. Options market makers send millions of quotes on options series to multiple exchanges throughout the trading day that allow institutional and retail investors to transact virtually any option quickly, fairly, and at reasonable prices. Encouraging market makers to add liquidity is essential to investor confidence, transparency, and price discovery.

Market makers make money by collecting the “spread” in an options trade. For example, a market maker may offer to buy a July 200 call for \$5.00 and sell the same option contract for \$5.05. The market maker will pocket the spread on the trade ($\$5.05 - \5.00 times a price multiplier, typically 100) and make a profit of \$5. Tighter spreads mean that buyers pay less for an option and sellers receive more, all else equal.

Because market makers are constantly executing trades in different options classes, they must manage risk across their portfolio. A key form of risk—known in the industry as “delta”—relates to the price of an underlying equity. Changes in equity prices drive changes in options prices and, because of the leveraged nature of options trades, can leave market makers exposed to enormous losses. Market makers

hedge against this risk by buying or selling the underlying stock, but their models must constantly update as the prices of thousands of underlying equities change throughout the trading day. The spread a market maker demands is directly related to the risk the market maker perceives when quoting an options contract; due to competitive pressures, market makers typically quote as tight as possible given risks and cost.

The problem of latency arbitrage arises because the dispersion of price information through electronic systems, while imperceptible to humans, is not instantaneous. Changes in equity prices on an exchange take time to propagate to market makers' systems, computer models take time to update, and new orders or cancels initiated by the market marker take time to flow back to the exchange. During this latency period—typically on the order of a fraction of a millisecond—market participants with faster access to new price information have a transient but significant advantage. A participant who knows that the price of an underlying equity is changing before the rest of the market does can “pick off” market makers by buying (or selling) just before the new information spreads and the prices of the options update accordingly. This results in an immediately losing trade for the market maker, who, if this occurs frequently enough, will likely widen his or her bid-ask spread over time to compensate, causing inferior fills for investors.

Citadel argues that using the term “latency arbitrage” to describe options trading based on movements in the underlying stock is improper because options markets are different from the equities, in which identical instruments trade on different exchanges. Pet. Br. 17–18. This disregards the well-known economics involved. Options are derivative instruments whose value is based directly on their underlying equities. In fact, many “deep in the money options” have strike prices sufficiently attractive relative to the underlying stock price that the options are nearly certain to be exercised. Such “100 delta” options exhibit a 100% price change correlation with the stock, and are often direct proxies for trading the stock itself, since the options will almost certainly be exercised and converted to shares of that underlying stock at expiry. An identical equity trading on another exchange is simply a different example of a correlated product—one that, like a deep in the money option, also happens to have a correlation of 100%.

Other options also exhibit quantifiable correlations to the underlying stock price. Describing the trading of options as categorically different than equities ignores that options prices are mathematically derived from the underlying stock price regardless of the degree of correlation. In other words, the latency arbitrage phenomenon is economically identical whether the reference price comes from the same instrument on another venue or from an underlying security of a listed option. In either case, the market maker faces the exact same problem: the exploitation of

speed differentials to trade against stale quotes in a correlated security. In fact, as the Commission recognized, the issue of latency arbitrage is more pronounced for options market makers than in other markets, if anything, because of the large number of options series a market maker must be able to quote. *See* 90 Fed. Reg. at 45870–71.

CTC can confirm that options market makers widen spreads today, and every day, to partially offset the effects of toxic order flow, including that driven by latency arbitrage, relative to what they would do if such toxic flow did not exist. Competitors may perhaps disagree about the magnitude of this effect, but claiming that latency arbitrage does not exist in the options market—*i.e.*, that market makers do not widen spreads due to delta pickoffs—is patently incorrect.

B. The IEX proposal would make markets fairer by reducing the potential for latency arbitrage.

Because market makers bake pickoffs into their required edge to quote, *any* mechanism that reduces assessed pickoffs will inevitably narrow quoted spreads to some degree. The IEX proposal approved by the Commission is one such mechanism. By introducing a *de minimis* speedbump that eliminates the advantage of the very fastest participants, the proposal would enable the exchange to intelligently revise or cancel market maker quotes in limited circumstances, reducing latency arbitrage costs for investor orders on IEX Options.

Reducing arbitrage costs, in turn, would reduce the disadvantage currently incurred on options exchanges by liquidity providers who are slower (by even the smallest margin) than the most aggressive liquidity takers and must spend vast sums on ever-greater technology infrastructure projects to keep up. Removing this disadvantage to liquidity providers stands to benefit investors through additional liquidity and tighter bid-ask spreads.

This conclusion is supported by peer-reviewed academic research. Professors Eric Budish, Peter Cramton, and John Shim, for example, have characterized the current situation as “a never-ending arms race for speed” and described the result as “a classic prisoner’s dilemma.” E. Budish et al., *The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response*, 130 Q.J. Econ. 1547, 1555 (Nov. 2015). Without a mechanism to mitigate latency arbitrage, arbitrageurs “invest in speed to try to win the race to snipe stale quotes” while liquidity providers must “invest in speed to try to get out of the way of the snipers.” *Id.* Trading firms “would be better off if they could collectively commit not to invest in speed, but it is in each firm’s private interest to invest.” *Id.* As these authors conclude: “sniping is negative for liquidity and that the speed race is socially wasteful.” *Id.* at 1594. In other words, technology expenditures often serve only to protect professional market participants from one another (and, perhaps, to generate a profit for purveyors of

market data and trading infrastructure) and do not fill any expressed need for additional speed on the part of end users.

To be sure, because there are two sides to every trade, there will always be winners and losers from disruption to the status quo. The pattern of comment letters filed in the underlying proceeding reveals the competitive dynamic. The largest, most entrenched U.S. equities options market making wholesalers (including Citadel) opposed the IEX filing, while smaller market makers—including CTC and at least one international firm that is a new entrant to U.S. options market making—supported it. This pattern suggests the proposal may encourage upstart competitors in the market making space, rather than entrenching incumbent advantages.

Ultimately, whether incumbents would prefer a regime that advantages those firms that can afford to spend the most to shave microseconds off their order latencies is beside the point. Just as the federal antitrust laws were enacted to protect competition, not competitors, *Atl. Richfield Co. v. USA Petroleum Co.*, 495 U.S. 328, 338 (1990), the Exchange Act was enacted to serve the public interest, not the interests of any particular group of participants. CTC agrees that exchanges are not and should not be viewed as “private clubs.” Pet. Br. 30 (quoting *Silver v. New York Stock Exchange*, 373 U.S. 341, 351 (1963)). Under 15 U.S.C. § 78f(b)(8), however, the relevant question in assessing whether a proposed rule’s burden on competition is “necessary or appropriate” is not whether one group stands to benefit from the

status quo. It is whether the marketplace as a whole will benefit if the rule is enacted. The Commission correctly concluded that the answer to this question was yes.

II. THE IEX PROPOSAL IS NOT UNFAIRLY DISCRIMINATORY.

Citadel and the *Amici* Exchanges also argue that the IEX proposal amounts to “unfair discrimination” Pet. Br. 30–39; *Amici* Exchanges Br. 13. In particular, Citadel alleges that it violates the Exchange Act to approve a rule that “gives IEX’s own market-making members a unique, at-execution repricing privilege and then forces all other brokers and investors to trade on those terms.” Pet. Br. 29; *see also id.* at 31. It is worth unpacking this criticism carefully, because it fails on two fronts.

A. Fair competition is not unfair discrimination.

Part of Citadel’s concern appears to be that IEX’s innovative structure would “unfairly” force market makers on other exchanges (here, somewhat confusingly called “brokers”) to also quote at better prices. If true, then investors trading on those markets would receive fills at those better prices—an excellent outcome for the investing public. Arguing that the IEX proposal is problematic because market makers will have to display better prices elsewhere is akin to suggesting Apple should not be allowed to add new features to the iPhone because doing so makes Google work harder. Competition that forces market-wide price improvement is precisely what the Exchange Act is designed to encourage.

Citadel’s poker analogy (Pet. Br. 32) is evocative but fundamentally mischaracterizes who the “favored players” are in today’s markets. A better analogy would be a card game where certain players have installed hidden cameras that let them see everyone else’s cards moments before each bet is finalized. These players have invested tens of millions of dollars in this system. When they spot that your hand is likely to beat theirs, they fold—instantaneously, before you can win anything from them. When they see your hand is weaker, they raise aggressively. Other players at the table know that this is happening. To compensate, they play more conservatively, making smaller bets and folding more often, because they know they will lose out on a material fraction of hands and take less on the hands they do win. IEX’s mechanism is like blurring the hidden camera feed somewhat—returning the game to a fairer contest where more players, including those who have not invested millions in a high-speed camera system, operate on similar information.

The similar “speedbump” in the equities markets demonstrates that there is no “unfair” discrimination in that analogous case. The SEC has already approved such delays on IEX and formerly on NYSE American. The DC Circuit upheld that determination. *See Citadel Secs. LLC v. SEC*, 45 F.4th 27 (D.C. Cir. 2022). Furthermore, while Citadel claims the options market operates under a “different framework than equities” (Pet. Br. 5), that is incorrect. *See pp. 8–9 supra*. Regardless

of technical differences, the fundamental principle that *de minimis* intentional delays are permissible under the securities laws applies equally here.

Acknowledging that the DC Circuit approved a *de minimis* delay in the equities context, Citadel later argues that there is an “irreconcilable” tension between the concept of a *de minimis* delay and IEX’s claims that the ORP will have a “substantial” impact on the marketplace. Pet. Br. 55 n.*. That is a non sequitur. Indeed, that the ORP entails only a *de minimis* delay is a virtue, not a vice. It shows that the IEX proposal is narrowly tailored to improve competition. A longer intentional delay would similarly reduce the risk of latency arbitrage, but may more significantly disrupt market participants’ ability to hedge and manage risk.

By using an exchange-controlled mechanism to mitigate pick-off risk that could otherwise be addressed only through massive technology expenditures, the IEX proposal is likely to lower technological barriers to entry and allow more firms, including those unwilling or unable to spend giant sums to build a microsecond-latency trading platform, to become competitive at providing liquidity. By helping to establish a market structure where massive technology expenditures are no longer as critical in preventing trading losses, and empowering existing liquidity providers to quote more aggressively, the IEX proposal will help foster a fairer and more open marketplace. That is the opposite of unfair discrimination.

B. The risk of missing a displayed quote has always existed.

Citadel also argues that the “protected” quotation rules in the Options Order Protection and Locked/Crossed Market Plan require brokers on other exchanges to route customer orders to the exchange with the best available price even though the quotes are not, in Citadel’s telling, “firm.” Pet. Br. 2. Although Citadel warns that the “prices displayed to investors” on IEX “will often be illusory,” *id.*, the reality is that orders sent to any electronic market have *always* been at risk of missing a fill. This may be due to a cancel that may already be in flight but not yet reflected in the published quote; an order update from another participant with a faster connection (for example, microwave versus fiber transmission); or a shorter geographical distance to the relevant exchange data center. Whatever the reason, IEX quotes would not be “stale” in any new or unique way.

The delay contemplated by the ORP is similar to delays inherent in many existing exchanges. The best national quotation for a given options contract (the “OPRA NBBO”), for instance, is aggregated and computed from the best bids and offers from all constituent options exchanges, which must then be transmitted to OPRA’s calculation engine at NYSE’s data center in Mahwah, New Jersey, and then transmitted nationwide. According to Nasdaq’s Chief Economist, the summed aggregation, calculation, and round-trip transmission times to (for example) Nasdaq’s data center in Carteret, NJ are about 500 microseconds, which already

exceeds IEX's proposed speedbump. Phil Mackintosh, Nasdaq, *Time is Relativity: What Physics Has to Say About Market Infrastructure* (Apr. 9, 2020), <https://www.nasdaq.com/articles/time-is-relativity-what-physics-has-to-say-about-market-infrastructure-2020-04-09>. The same publication noted that even a delay longer than IEX's proposed 350 microseconds would have "no impact on human traders or professionals watching a screen." *Id.*

Likewise, in 2017, NYSE defended a proposed speedbump on its own equities exchange (later approved by the Commission) in part by stating that a 350 microsecond delay was "well within the geographic and technological latencies experienced by market participants when routing orders" and "would provide Exchange systems sufficient time to update prices based on market data disseminated by other markets." Letter from E. King (NYSE) to B. Fields (SEC) dated May 11, 2017, available at <https://www.sec.gov/comments/sr-nysemkt-2017-05/nysemkt201705-1748553-151659.pdf>. The geographic latencies NYSE referenced are identical between equities and options markets (since they are operated by many of the same exchange groups in the same data centers) and have not changed since 2017. Indeed, given that the speed of light in fiber is a constant, they will never change, barring a physical relocation of an exchange data center. This directly contradicts any suggestion that the speedbump poses unprecedented harm to market participants.

It has also been suggested that “jitter”—or random delays in trade matching due to technical limitations—is distinguishable from an “intentional” speedbump. In CTC’s view, this is an artificial distinction. Jitter in exchange matching engines and market data technology routinely exceeds the duration of the proposed speedbump. As an example, since 2021, the difference between the reported OPRA 10th and 99th percentile latencies has ranged from hundreds of microseconds to over 2,870 microseconds. OPRA SIP Metrics Feb. 2024, <https://cdn.opraplan.com/documents/OPRA-SIP-Metrics-February-2024.pdf>. Exchanges often choose not to address this jitter immediately to save money on infrastructure costs or due to other technical priorities, meaning such latency could more accurately be considered intentional latency through inaction. The idea that this latency, quietly left in place for years, is perfectly acceptable whereas IEX’s fully-disclosed speedbump, executed in hardware and described in a detailed regulatory filing, is unacceptably novel is not a coherent distinction.

Stale quotes do not arise only from the physical and geographic infrastructure of the markets. Often, akin to the ORP proposed by IEX, they are triggered intentionally in the course of normal exchange operations. Market maker risk controls (MMRCs), for example, have been a standard feature of options exchanges for approximately twenty years. MMRCs are mechanisms implemented by options exchanges that allow the exchanges, on their own initiative, to cancel a market

maker's displayed quotes in response to certain events without an explicit, contemporaneous instruction from the market maker. Triggers for these cancels may include trading a prescribed number of contracts, executions, or complete quote fills. The relevant parameters are set by market makers within ranges allowed by the exchanges pursuant to SEC-approved rules.³

Consider the following example. A market maker in Apple options has configured MMRCs to pull all outstanding quotes on the Nasdaq ISE exchange after trading a total of 50 contracts (an SEC-approved feature which has been available on ISE since approximately 2001).

Time	Event
T+0	Nasdaq ISE receives the market maker's bid of \$6.00 for 50 July 200 calls
T+1	Nasdaq ISE receives the market maker's bid of \$4.00 for 50 July 205 calls.
T+2	Retail Customer A sends an order to sell 50 AAPL July 200 calls at \$6.00
T+3	Retail Customer B sends an order to sell 50 AAPL July 205 calls at \$4.00

³ In addition to MMRCs, multiple exchanges use SEC-approved mechanisms that reference equity trades or prices, including protections against selling options under parity; Obvious Error rules that use the underlying price to come up with a theoretical price for a given option for purposes of making rulings to bust or adjust trades; and automatically halting options markets (*i.e.*, making previously displayed quotes suddenly inaccessible) when the underlying equity price has entered a Limit Up/Limit Down Volatility Trading Pause.

T+4	Nasdaq ISE receives and executes Customer A's order, and the MMRC is tripped, cancelling all of the market maker's AAPL quotes
T+5	Nasdaq ISE receives Customer B's order

In this example, even though a displayed bid of \$4.00 was available when Customer B's order was sent, Customer B will not receive a fill because when the exchange receives the order, the quote that the customer was trying to hit will already have been canceled by the MMRC.

Similar to IEX's proposal, the intent of MMRCs is to foster greater liquidity provision, and therefore better prices for investors, by allowing market makers to quote in confidence that they won't be "run over" and simultaneously forced into a large number of risky, correlated trades on a sharp market move. To CTC's knowledge, moreover, these longstanding MMRC mechanisms are uncontroversial.

MMRC mechanisms also share the precise features that many commentators (including Citadel) claimed make the IEX proposal uniquely problematic. First, an MMRC may cancel a quote in one options series without the exchange ever having executed any order in that particular option series. Typically, MMRCs are configured to operate across all instruments in a given options class, so a fill in one series can trigger cancellation of quotes in an entirely different series that was never traded. That is why, in the example above, the order for Apple July 205 calls was

canceled even though the order triggering the MMRC was for a July call at a different strike price.

Second, MMRC cancellations are typically initiated atomically and in sequence directly by the exchange matching engine. This means that they “jump the queue” and result in cancellations that prevent execution of otherwise-marketable orders that may already be in flight to the exchange, or even already queued up for execution in the exchange matching engine. That is why an MMRC triggered at time T+4 resulted in the cancellation of an order sent at time T+1 for which a matching order has already been received.

These are desirable and necessary features of MMRCs, which the SEC has approved and which are offered in various forms on all major options exchanges. It is unclear how exchanges using these controls “honor displayed quotes” in some way that the IEX proposal does not. Citadel claims that certain risk controls, like “purge ports,” are different because they “prevent errors before execution and never relieve firms of their obligation to honor displayed quotes.” Pet. Br. 22–23. But that is incorrect. The ORP also operates before execution, and the implication that market makers trading on exchanges always honor displayed quotes at the time an order is sent is wrong. In the example above, for instance, Customer B’s order was sent to a displayed quote that vanished before the order could be executed. That is precisely the scenario Citadel claims is unprecedented.

In sum, the IEX proposal would promote fair competition and would not fundamentally change the risk of a missed fill for investors sending orders to exchanges. The Commission was right to approve it.

CONCLUSION

The petition for review should be denied.

Dated: February 6, 2026

Respectfully submitted,

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CERTIFICATE OF COMPLIANCE

I, Jed W. Glickstein, hereby certify that the Brief of CTC, LLC as *Amicus Curiae* in Support of Respondent complies with the type-volume limitations of Federal Rules of Appellate Procedure 29(a)(5) because it contains 4,715 words, excluding the portions of the response exempted by Federal Rule of Appellate Procedure 32(f). In preparing this certificate, I relied on the word count tool of the word-processing system used to prepare the brief, Microsoft Word 365.

I further certify that the brief complies with the typeface and type style requirements of Federal Rule of Appellate Procedure 32(a)(5) and (6) because it has been prepared in a proportionally spaced typeface using Microsoft Word 365 in Times New Roman 14-point font.

Dated: February 6, 2026

Respectfully submitted,

/s/ Jed W. Glickstein

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CERTIFICATE OF SERVICE

I, Jed W. Glickstein, hereby certify that I served the Brief of CTC, LLC as *Amicus Curiae* in Support of Respondent on February 6, 2026 using the CM/ECF system, which effected service on counsel of record for all parties.

Dated: February 6, 2026

Respectfully submitted,

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